Lateral analysis - Level 2 - First dry critical speed calculation

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1 Purpose.

This document is related to the dynamic analysis requirements for VS rotordynamic pump, especially to lateral analysis.

The procedure to define what level of detail for the dynamic analysis is required to be in accordance with good practice of VS pump design is described in G.10.00.09 and *ANSI/HI 9.6.8-2014*: if calculated RUN is higher than 20 and less equal to 50, a **Level 2 – Intermediate method - Lateral rotordynamic analysis – First dry critical speed calculation** is required.

This document describes how to perform an undamped natural frequency (lateral) analysis for a vertical pump.

2 Abbreviations.

VS	vertically suspended (API610)
OEM	original equipment manufacturer
DOF	degree of freedom
FEA	finite element analysis
OH	overhung (API610)
BB	between bearing (API610)
R	risk number (ANSI HI 9.6.8-2014)
RUN	R multiplied by U (ANSI HI 9.6.8-2014)
U	uncertainty value (ANSI HI 9.6.8-2014)
ROT_x	rotational degree of freedom around x axis in cartesian coordinate system
ROT_y	rotational degree of freedom around y axis in cartesian coordinate system
ROT_z	rotational degree of freedom around z axis in cartesian coordinate system
U_x	translational degree of freedom in x direction in cartesian coordinate system
U_y	translational degree of freedom in y direction in cartesian coordinate system
U_z	translational degree of freedom in z direction in cartesian coordinate system

3 Scope.

This procedure is applicable to VS pumps.

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4 General procedure: description.

Lateral rotordynamic analyses for VS pumps are different from those analyses done for OH or BB pumps. For the purpose of this discussion, between bearing/single stage centrifugal pumps that can be mounted horizontally or vertically are analysed as if in horizontal position.

In case of a VS pump, the type of computational lateral analysis used for OH or BB pump is often not useful since the bearings and the interaction with the relatively low-stiffness support structure are quite flexible and nonlinear in the reaction to load, and the stiffness and damping can be difficult to accurately predict due to a lack of well-defined hydrodynamic fluid film being formed. However, to determine if the bearing span is sized properly, OEMs typically perform a basic natural frequency calculation assuming the line shaft is a simply supported beam with pin support at the line shaft bearings. Fortunately, the line-shafting of VS pump is designed with long L over D ratio and high clearance, line-shaft bearing, such that the line shafting tends to act as a violin string as the musician's fingers move to touch different locations. In this manner, each rotor's natural frequency is a constantly "moving target", making significant resonance very unlikely. Therefore, a more practical use of vibration prediction resources is typically the vertical pump's structural natural frequency, particularly the aboveground "reed frequency" mentioned in.

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5 Methodology, analysis and interpretation of results.

Methodology.

Software/Tools:

- general purpose FEA programs or specialty rotordynamic programs;
- software should account for gyroscopic effects.

The tool described in this procedure is:

- DyRoBes

Model.

VS pumps with thrust bearing: mass-elastic model of pump rotor only. Impeller and coupling half are considered as lumped mass effect (mass and inertia) without stiffness contribution.

VS pumps without thrust bearing: mass-elastic model of motor – coupling - pump shaft train (high lateral stiffness of rigid coupling does not laterally decouple pump rotor from motor rotor).

Rigid coupling modelled with beam element.

In some cases, introduction of zero-length coupling elements (linear spring element) at coupling hub locations is necessary to simulate the effect of loose-fit coupling hubs.

Shaft segments modelled with beam or pipe elements; concentrated masses modelled with lumped mass elements.

Elements have four DOF:

- translations in both lateral directions (horizontal U_x and vertical U_y; axial U_z is constrained at the trust bearing) and rotations around lateral axes (ROT_x and ROT_y; torsional ROT_z is constrained at all nodes).

Shaft sleeves, rotating portion of the mechanical seal, etc. are considered as a distributed mass effect (no stiffness contribution).

Interaction forces from annular seals and impeller/casing interaction are not considered (only liquid effects are from fluids contained within impeller passageways, and fluids displaced by wetted component volume).

Radial bearings and line-shaft bearings are the only rotor support.

All bearing stiffness and bearing support stiffness are infinite (rigid bearings).

Non-linear effects of vertical shaft axial loading by gravity (“guitar string” and “pendulum effect”) are not included.

Analysis.

Undamped natural frequency analysis (modal).

Rotor speeds: minimum, rated and maximum continuous operating speed (one speed in case of fixed speed application).

Interpretation of results.

A Campbell diagram is typically not strictly necessary for this type of analysis.

Results of the natural frequency analysis (solution of the eigenvalue problem) are sufficient: this includes mode shape (eigenvector) and associated natural frequency (eigenvalue).

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6 Acceptance Criteria.

Shape and associated natural frequency of the first mode is relevant.

In case the first dry natural frequency is at least 20% above the maximum continuous operating speed, then the rotor is expected to have no lateral rotordynamic issue with 1X synchronous speed excitation during normal (wet) operation.

This dry critical speed analysis cannot be used to judge the stability of the pump rotor, nor its degree of forced response ((i.e., the vibration amplitudes expected).

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7 Modelling recommendations.

7.1 Shaft modelling.

One of the most common questions about shaft modelling is how many elements or stations are required to get good results.

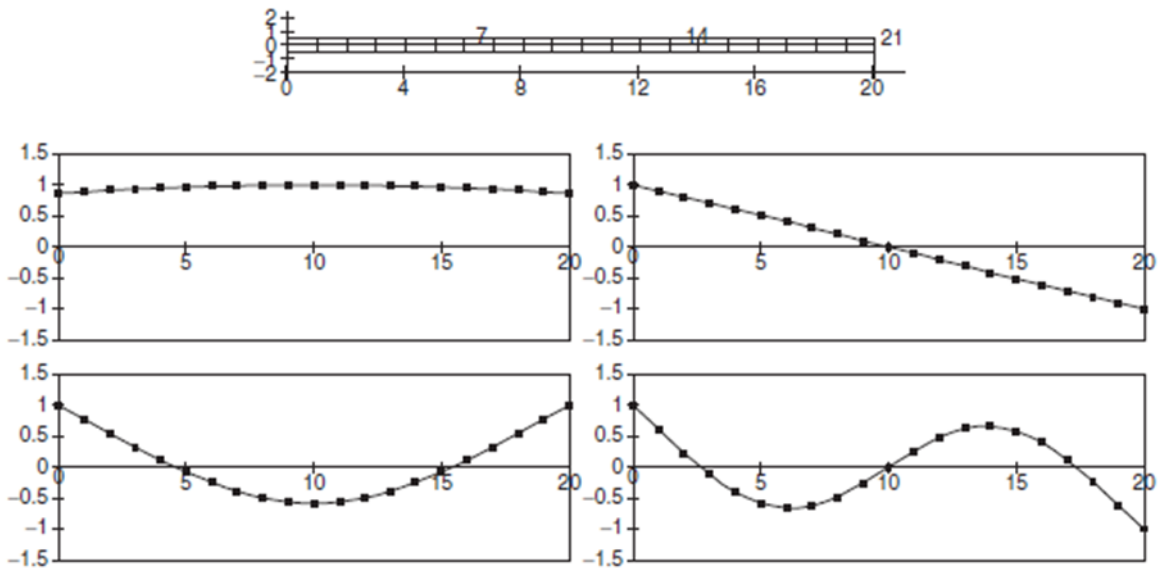

Figure 1 Model of uniform beam used for mode accuracy study, and mode shapes of the first four normal modes.

Figure 1 shows a uniform steel beam 20 in. long, 1 in. in diameter, and with an undamped bearing at each the end having K = 1000 lb/in. The mode shapes of the first four modes are also shown. The first two modes are fairly rigid. Modes like these don't require many elements for good accuracy. The second two modes look like a half sine wave and a full sine wave.

Figure 2 shows that accuracy depends not only on the number of elements, but also on the formulation of the elements. Notice in Figure 2 that the rigid modes have very small percentage errors. This is because there is little to no flexure of the elements. The flexible modes, however, illustrate the effect of the number of elements and also the difference between the four different element types. For our uniform beam test rotor, the four-element case corresponds to an L/D of 5. For this case the first mode, which is a rigid shaft mode, has frequency errors less than 1 percent. Even the second mode, which is also a rigid shaft mode, has errors less than 1 percent for all but the uncorrected 2-node element. For the first flexible mode, however, four elements are not enough, and the errors exceed 10 percent. Based on the errors computed for this example, a guideline of "use elements with L/D < 1" will result in acceptable accuracy for all cases considered.

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Another reason to maintain an element L/D < 1 is that when using a transfer matrix-based calculation procedure, the calculations can become numerically ill-conditioned if the L/D of very many elements is much longer than one. This is an aspect specific to the algorithm, and not the rotor system model.

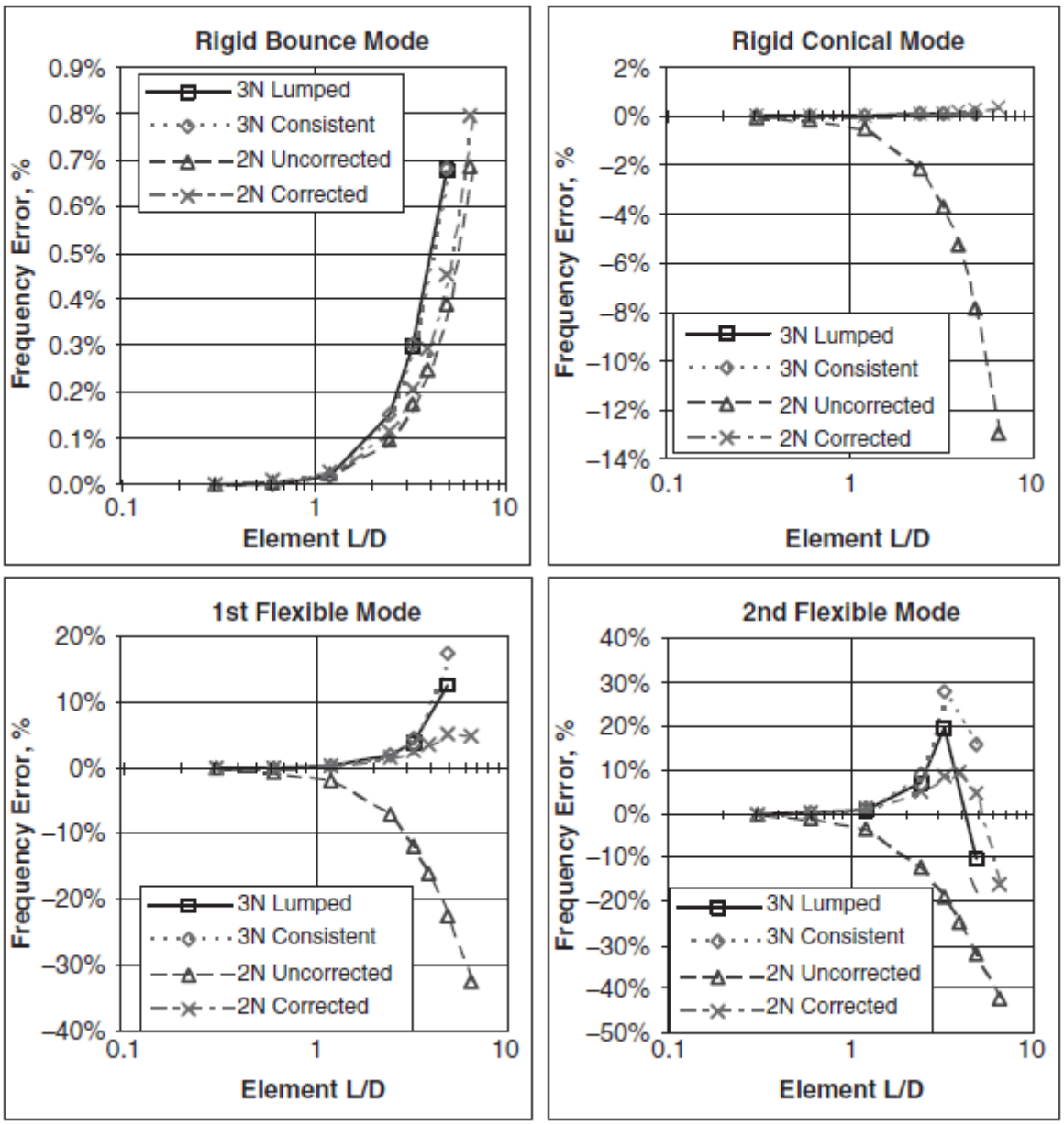

Figure 2 Mode accuracy versus number of elements for four different element definitions.

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7.2 45 degrees rule

A common situation encountered in rotor modelling is a large abrupt change in diameter. The bending stiffness of the shaft is not able to change quite as sharply as the diameter. A step change in diameter of one-third or more is considered abrupt. The material outside the 45° line shown in Figure 3 will not contribute significantly to the bending stiffness of the section. Two ways to model this section are shown, one using cylindrical beams and one using conical beams. Both approaches model the transition section with a layered pair of beams (inner and outer layers). The inner beam has the actual density and modulus of the shaft material. The outer beam has the actual density, but zero modulus so as to not contribute any stiffness.

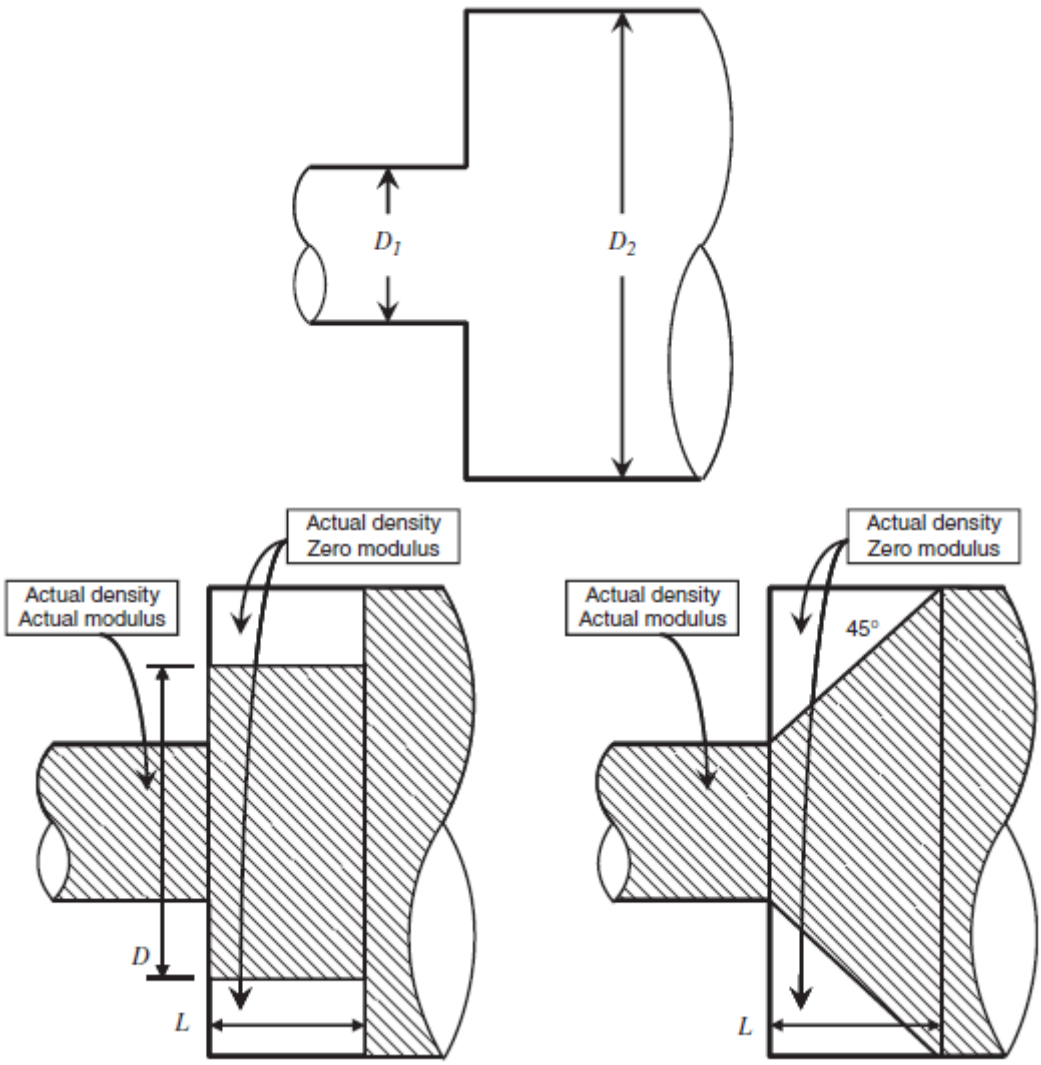

Figure 3 Two methods of the 45-degree rule for modelling abrupt diameter changes.

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7.3 Stations for Added Weight.

The inertia properties of pump impellers are most often added to the rotor shaft model as added concentrated weights and inertias. When constructing the model of the underlying shaft, be sure to provide stations at or close to the centre of gravity of each impeller. These will be the locations where the concentrated masses will be placed.

7.4 Fluid "Added Mass".

The fluid surrounding the rotor adds inertia to the rotor in two ways:

- the fluid trapped in the impeller passages adds mass directly, and this can be calculated based on the volume in the impeller passages times the pumped fluid density;
- the fluid around the periphery of the impellers that is displaced by the vibrating motion of the impeller; this part of the added mass is equal to the "swept volume" of the impellers and immersed shafting, times the density of the pumped liquid.

References

Turbomachinery Rotordynamics – Phenomena, Modelling & Analysis – Dara Childs

Machinery Vibration & Rotordynamics – John Vance - Fouad Zeidan - Brian Murphy

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API PUBBLICATION 684, 1st Edition, February 1996

ANSI/HI 9.6.8-2014 Rotordynamic Pumps – Guideline for Dynamic Pumping Machinery

G.10.00.09 - Guidelines for dynamic analysis complexity level

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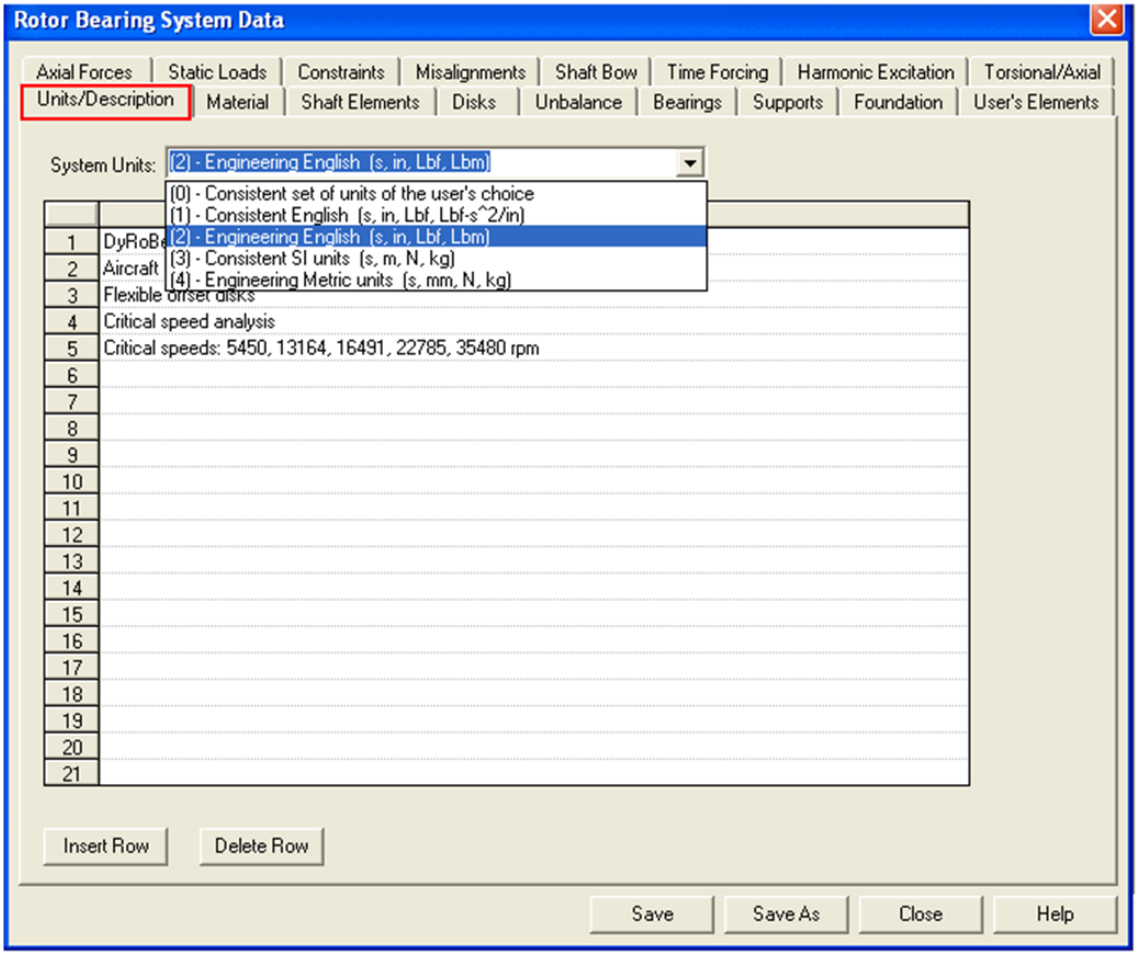
Annex

Annex 1 Uses of DyRoBes.



Step1: unit system.

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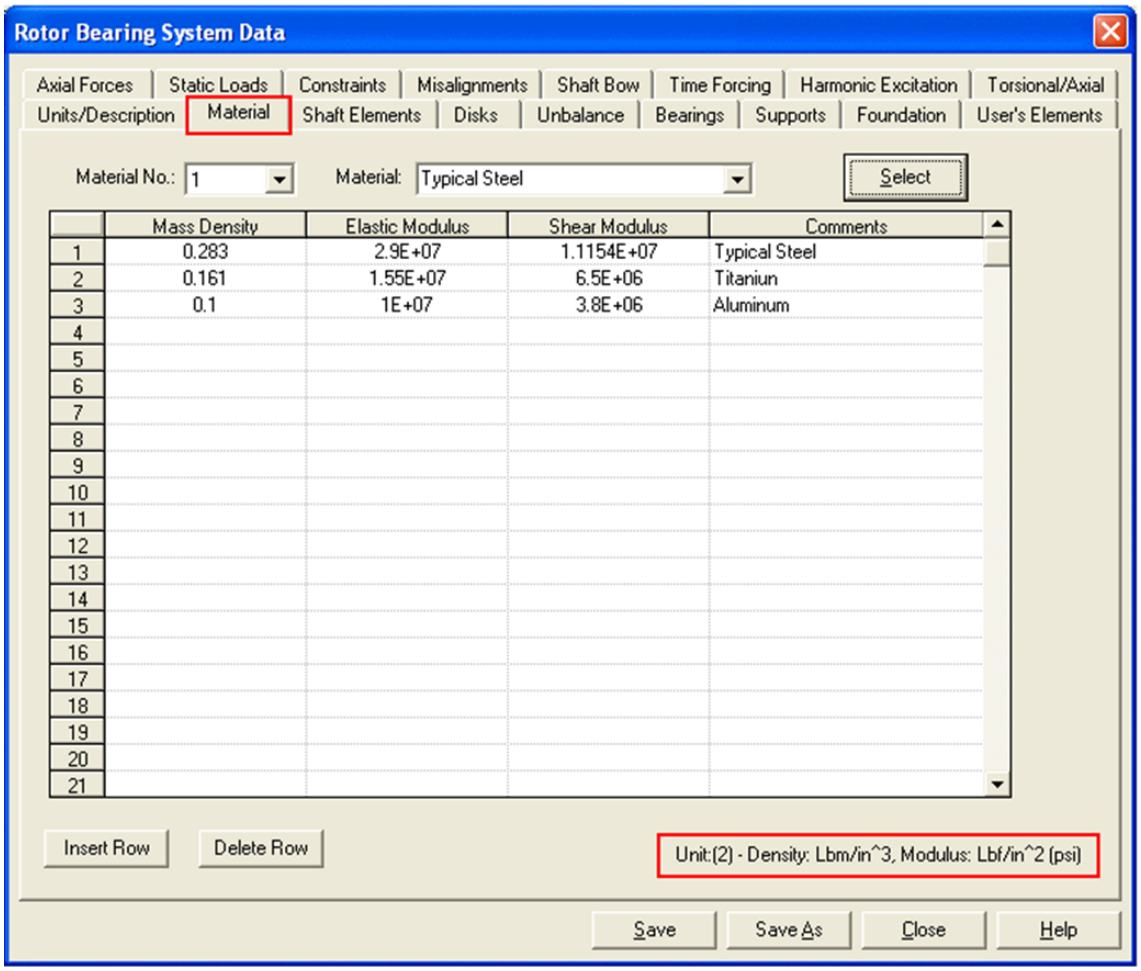

Step 2: material.

Enter the density, elastic and shear modulus of the materials of the shaft. Include a “dummy” material with very low elastic and shear modulus to model the rotors parts with no stiffness contribution.

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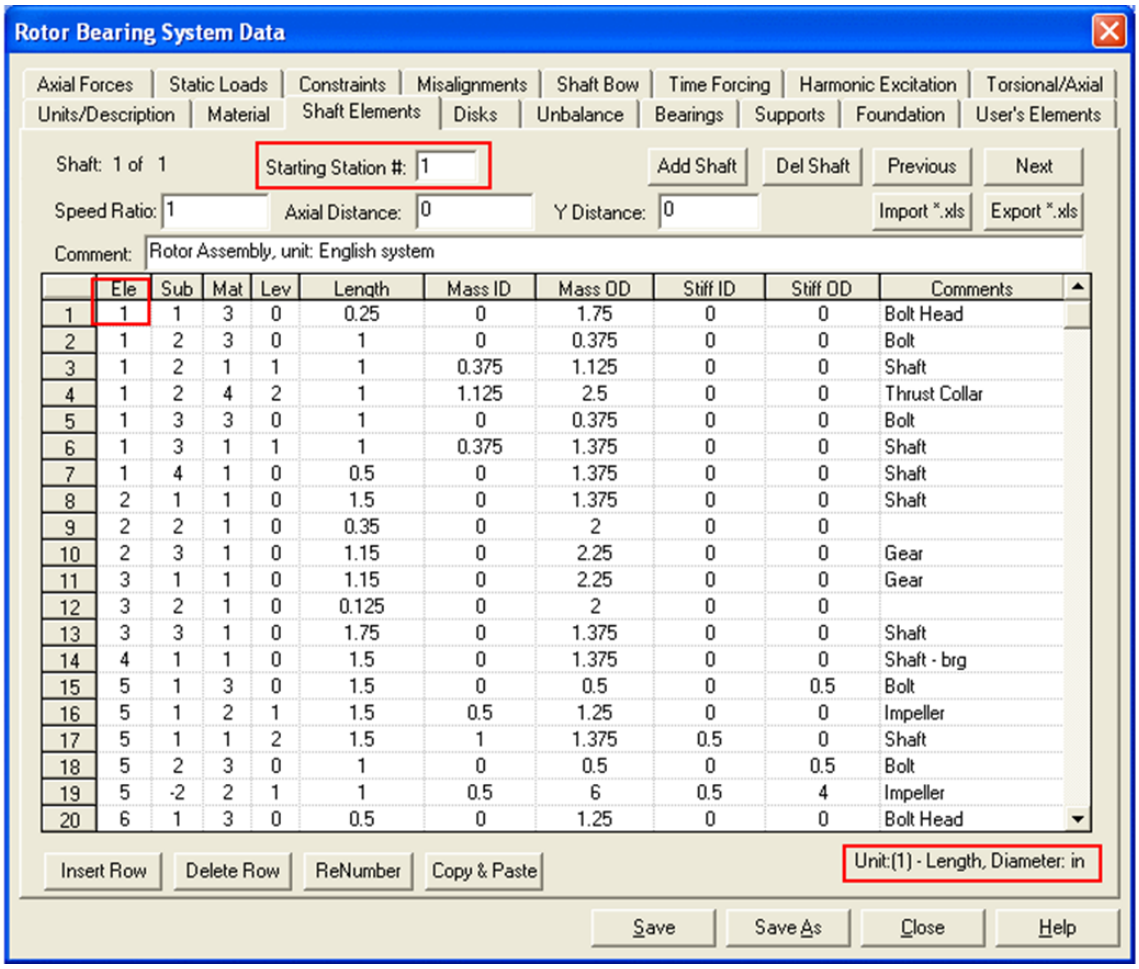

	Ele	Sub	Mat	Lev	Length	Mass ID	Mass OD	Stiff ID	Stiff OD	Comments
1	1	1	3	0	0.25	0	1.75	0	0	Bolt Head
2	1	2	3	0	1	0	0.375	0	0	Bolt
3	1	2	1	1	1	0.375	1.125	0	0	Shaft
4	1	2	4	2	1	1.125	2.5	0	0	Thrust Collar
5	1	3	3	0	1	0	0.375	0	0	Bolt
6	1	3	1	1	1	0.375	1.375	0	0	Shaft
7	1	4	1	0	0.5	0	1.375	0	0	Shaft
8	2	1	1	0	1.5	0	1.375	0	0	Shaft
9	2	2	1	0	0.35	0	2	0	0	
10	2	3	1	0	1.15	0	2.25	0	0	Gear
11	3	1	1	0	1.15	0	2.25	0	0	Gear
12	3	2	1	0	0.125	0	2	0	0	
13	3	3	1	0	1.75	0	1.375	0	0	Shaft
14	4	1	1	0	1.5	0	1.375	0	0	Shaft - brg
15	5	1	3	0	1.5	0	0.5	0	0.5	Bolt
16	5	1	2	1	1.5	0.5	1.25	0	0	Impeller
17	5	1	1	2	1.5	1	1.375	0.5	0	Shaft
18	5	2	3	0	1	0	0.5	0	0.5	Bolt
19	5	-2	2	1	1	0.5	6	0.5	4	Impeller
20	6	1	3	0	0.5	0	1.25	0	0	Bolt Head

Step 3: shaft elements

The shafts of the rotor system are numbered consecutively from 1 to Ns and the finite element stations of the model are numbered consecutively starting with 1 at the left end of shaft 1 and continuing to the last station at the right end of shaft Ns. The shafts are made up of Elements with the numbering for each shaft starts at the left end. Stations are located at the ends of the Elements. The term Station is commonly used in the rotordynamics instead of the term Node which is generally used in finite element literature because of the alternate meaning that Node has in the vibration mode shapes. Two types of elements are included in DyRoBeS:

- cylindrical Element;
- conical (Tapered) Element.

Element *i* is located immediately to the right of station i. Each element may possess several subelements (starting with 1 at the left of the element) thereby allowing for reasonable flexibility in modelling systems with several geometric discontinuities. Each subelement may possess several levels or layers (starting with zero). When using subelements, it should be kept in mind that Disks and Bearings can only be placed at the ends of the elements (stations).

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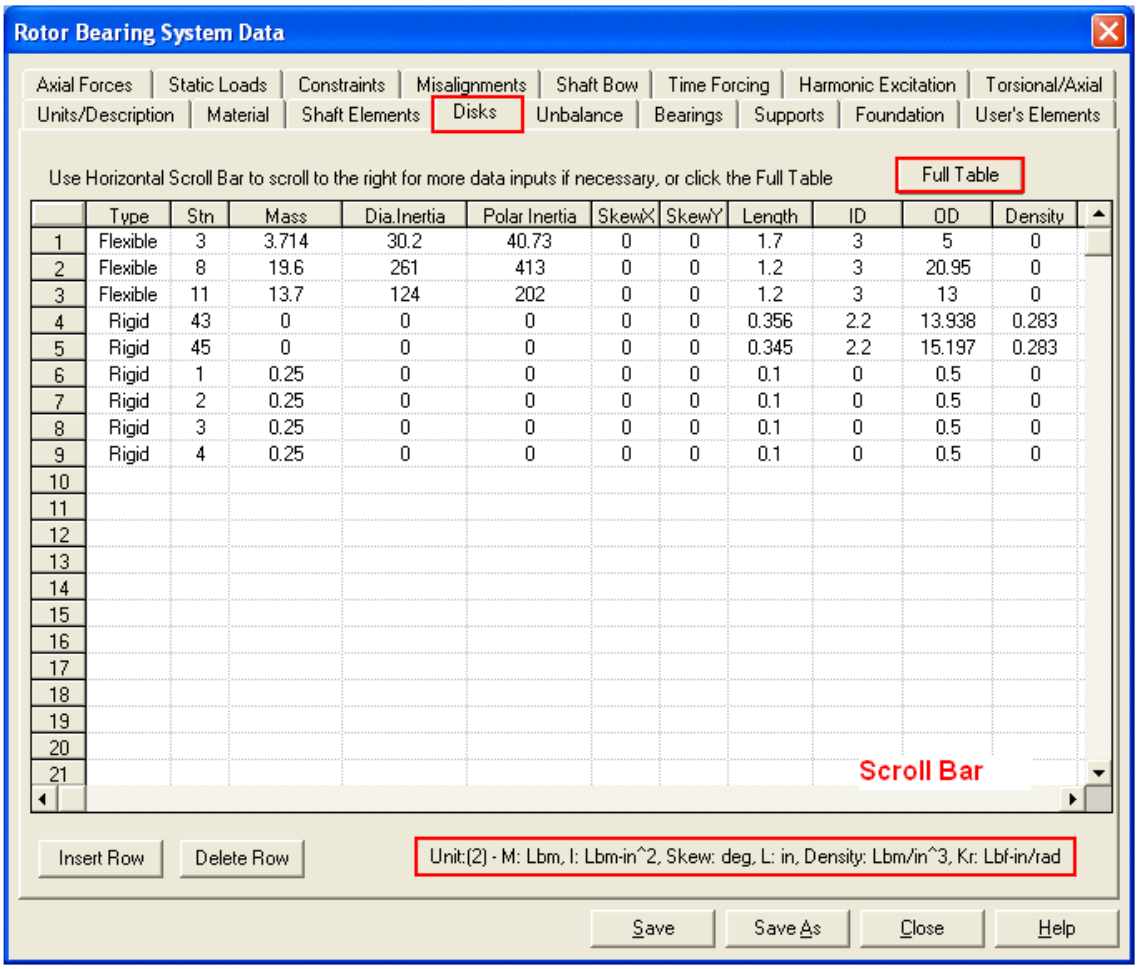

	Type	Stn	Mass	Dia.Inertia	Polar Inertia	SkewX	SkewY	Length	ID	OD	Density
1	Flexible	3	3.714	30.2	40.73	0	0	1.7	3	5	0
2	Flexible	8	19.6	261	413	0	0	1.2	3	20.95	0
3	Flexible	11	13.7	124	202	0	0	1.2	3	13	0
4	Rigid	43	0	0	0	0	0	0.356	2.2	13.938	0.283
5	Rigid	45	0	0	0	0	0	0.345	2.2	15.197	0.283
6	Rigid	1	0.25	0	0	0	0	0.1	0	0.5	0
7	Rigid	2	0.25	0	0	0	0	0.1	0	0.5	0
8	Rigid	3	0.25	0	0	0	0	0.1	0	0.5	0
9	Rigid	4	0.25	0	0	0	0	0.1	0	0.5	0
10											
11											
12											
13											
14											
15											
16											
17											
18											
19											
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21											

Step 4: disc.
A concentrated disk can be located at any finite element station. Disks cannot be placed at a substation, and must be placed at stations. Multiple disks are allowed at the same station.

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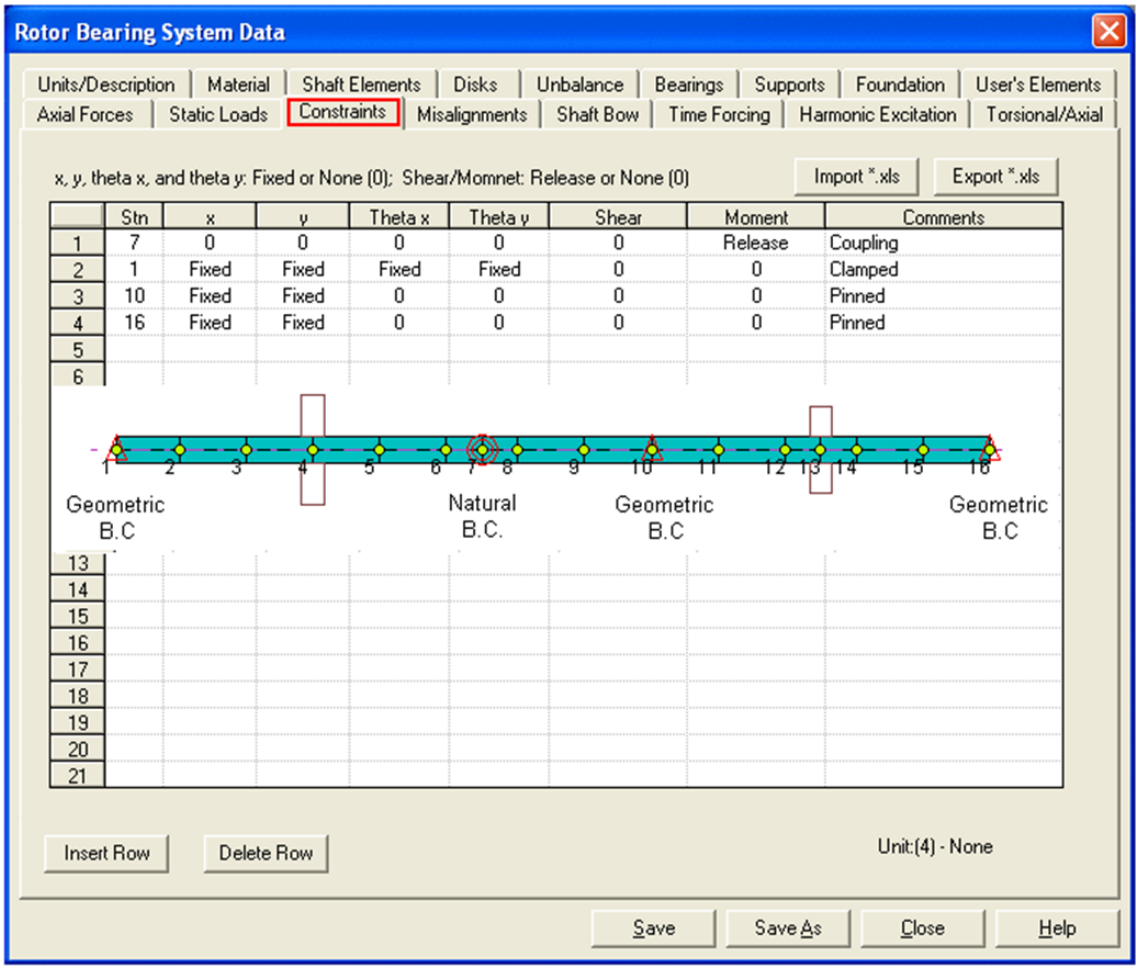

	Stn	x	y	Theta x	Theta y	Shear	Moment	Comments
1	7	0	0	0	0	0	Release	Coupling
2	1	Fixed	Fixed	Fixed	Fixed	0	0	Clamped
3	10	Fixed	Fixed	0	0	0	0	Pinned
4	16	Fixed	Fixed	0	0	0	0	Pinned
5								
6								
13								
14								
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Step 5: constraints.
Add constraint (fixed X and Y) to all bearing stations.

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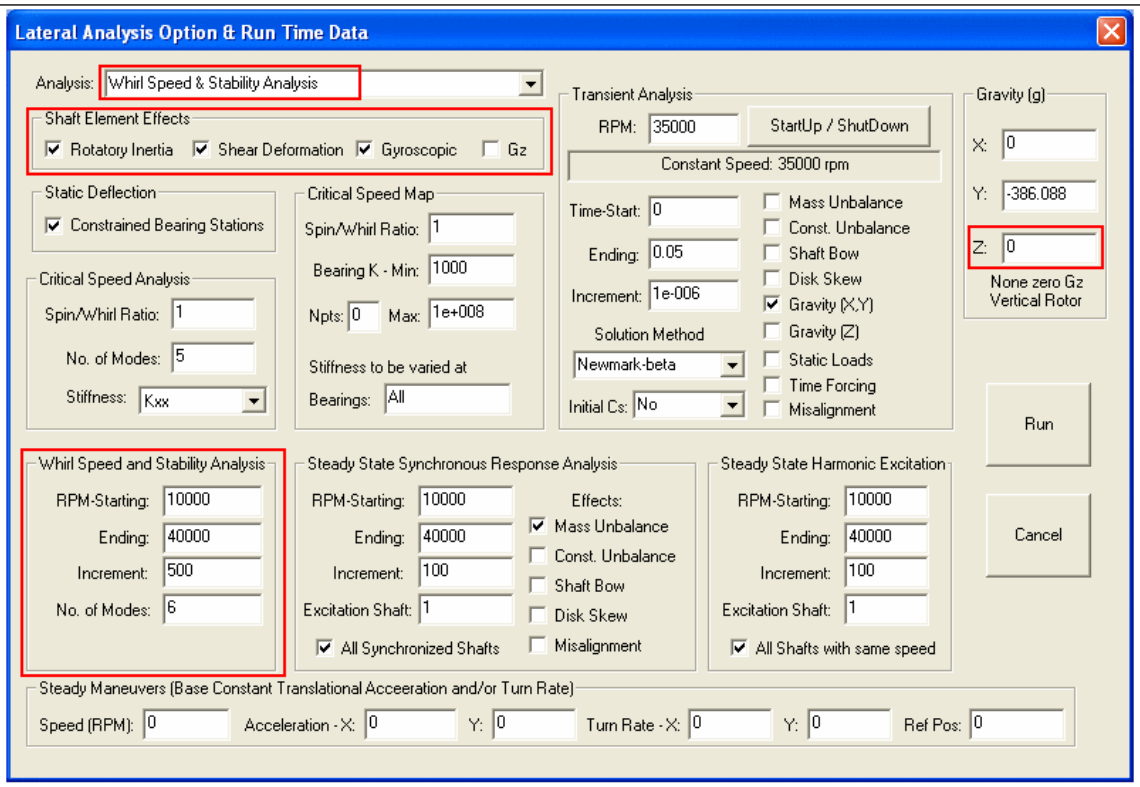
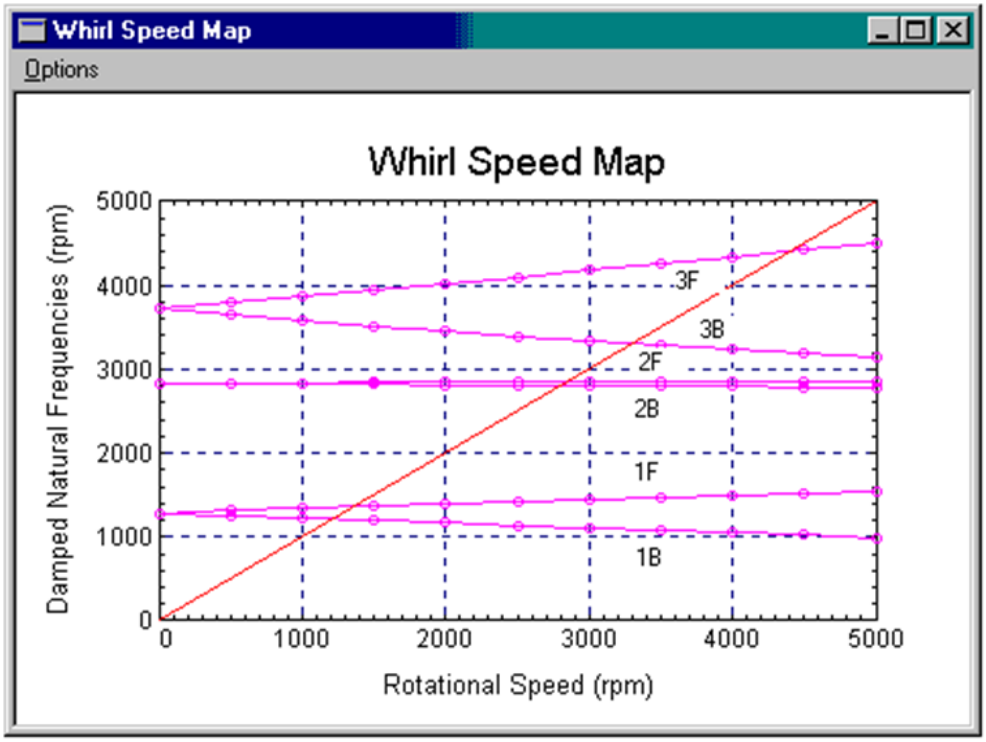


Step 6: analysis.

Complete a Whirl Speed & Stability Analysis:

- RPM-Starting: 0% of maximum speed;
- ending: greater than 120% of maximum speed;
- number of Modes: at least two



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Step 7: postprocessing.
Create a Whirl Speed Map.
The excitation slope must be 1.0 (red line); it is defined under “Options” in the graph.
If all the intersection between red line (synchronous excitations) and pink lines (natural frequencies) has an abscissa greater than 120% of maximum speed), the pump is verified

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